Structure-Aware Feature Refinement for Fine-Grained Industrial Anomaly Segmentation

Track – Industrial Track: Regular Setting

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Abstract

Robust unsupervised anomaly detection (UAD) in real-world industrial scenarios remains a challenging task. While recent prototype-based methods achieve strong results on traditional benchmarks, they exhibit clear performance degradation on the MVTec AD 2 benchmark due to its high-resolution imagery, small-scale defects, and large lighting shifts. To address these challenges, we propose a robust prototype-based segmentation framework with three key components: (1) strong self-supervised DINOv3 ViT-L/16 encoder for feature representations that are stable under photometric variation; (2) ***Graph Attention Context Block*** *after the prototype bottleneck, which lets normal prototype tokens exchange information through learned relational structure to cover the heterogeneous appearance patterns required by a class-agnostic pipeline; and (3)* ***Local Context Block*** *at each decoder level, which restores the local spatial inductive bias that pure transformer decoders lack and improves the segmentation of small defects. At inference time, anomaly maps are computed from the cosine distance between encoder and decoder features. On the MVTec AD 2 dataset, our method achieves XX.XX% SegF1 on $TEST_{priv}$ and XX.XX% SegF1 on $TEST_{priv,mix}$ under the regular setting. Code is available at https://github.com/WKlee0607/isvl_sj.*

1. Introduction

1.1. Background

Unsupervised anomaly detection (UAD) is a foundational task in industrial visual inspection. In this setting, models are trained only on defect-free images and must identify deviations from normality at inference time. The one-class paradigm has become standard because the set of possible defects is open-ended and rarely available in advance. Driven by this practical relevance, recent methods—such as PatchCore [7] and EfficientAD [1] based on feature embedding, Reverse Distillation [3, 9] based on reconstruction, and more recent prototype-based methods such as CPR [5] and INP-Former [6]—now report near-saturated performance on traditional benchmarks like MVTec AD [2], frequently exceeding 99% image-level AUROC.

However, this apparent saturation hides a large gap with real-world deployment. Production environments introduce distribution shifts that controlled benchmarks do not capture, including changing lighting conditions, sensor noise, transparent or reflective materials, and large intra-class appearance variation. Under these shifts, pixel-level localization—which is what factory inspection actually requires—degrades much more severely than image-level scores suggest. Addressing this gap requires UAD methods that remain reliable under real-world variation, rather than methods that only excel on clean benchmark distributions.

1.2. Challenge Description

The VAND 4.0 Industrial Track is built around the MVTec AD 2 dataset [4], which is designed to expose this benchmark-to-deployment gap. The dataset contains eight high-resolution industrial scenarios—*can*, *fabric*, *fruit jelly*, *rice*, *sheet metal*, *vial*, *wall plugs*, and *walnuts*—each captured under regular lighting for training and under a mixture of seen and unseen lighting conditions for testing. Images are high-resolution and often contain anomalies that occupy only a few hundred pixels, which requires the model to handle both global context and fine local detail at the same time.

The evaluation protocol provides three test splits: a small public split for development, a private split that shares the training-domain lighting, and a mixed private split that combines seen and unseen lighting conditions. The official ranking is the average rank of mean pixel-level SegF1 across the two private splits, and ties are broken by the

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absolute SegF1 gap between them. This directly rewards methods whose performance is *stable* across lighting domains rather than methods that excel on a single distribution. The challenge also imposes two deployment-realistic constraints on the model design. First, the pipeline must be class-agnostic: the same architecture and hyperparameters must handle all eight categories, so it is not allowed to pick a specialist model per object. Second, SegF1 requires committing to a single decision threshold without access to anomalous validation data. In addition to general performance, the challenge introduces a separate efficiency track that jointly scores accuracy, throughput, memory, and power consumption.

2. Methodology

2.1. Model Design

2.1.1. Approach

Our framework consists of three sequential stages: data preprocessing, anomaly detection, and postprocessing. The anomaly detection stage is built upon INP-Former [6], a prototype-guided feature reconstruction method that extracts Intrinsic Normal Prototypes (INPs) directly from the input image. We retain the core architecture of INP-Former—comprising the INP extractor, bottleneck, and INP-guided decoder—and introduce a set of targeted modifications to improve anomaly localization under challenging industrial conditions.

Data Preprocessing. Rather than adopting a multi-scale image pyramid, we employ a fixed grid-based splitting strategy in which each high-resolution input is partitioned into a 4×4 grid of sixteen non-overlapping sub-images. Each sub-image is treated as an independent sample throughout training and inference, effectively increasing the local resolution after resizing and enabling the model to capture small defects and fine-grained anomalies that would otherwise be suppressed when the full image is directly downsampled to the network input size. During inference, the resulting sub-image anomaly maps are reassembled according to their grid indices to reconstruct the full-image layout.

Anomaly Detection. We adopt INP-Former as the base reconstruction framework, training exclusively on normal samples. The model learns to reconstruct normal feature representations via intrinsic normal prototypes; at test time, anomalous regions produce elevated discrepancies between encoder and decoder features, which serve as the anomaly score. To strengthen the representation of local normal structures, we introduce a graph attention context module immediately after the INP bottleneck. Patch tokens are treated as graph nodes, with edges defined between spatially adjacent patches. The module refines each patch representation by aggregating features from its local neighborhood, enabling the model to capture normality not only from individual patch appearances but also from the structural relationships among neighboring patches.

We further augment the decoder with convolutional local-context refinement applied at every decoder level. This mechanism improves spatial consistency in the reconstructed features and preserves fine structural details around small or thin anomalous regions. To enhance generalization, we apply data augmentation comprising random horizontal flipping and brightness-based ColorJitter, the latter serving as a lighting augmentation strategy to improve robustness against the illumination variations commonly encountered in industrial inspection settings.

Postprocessing. Continuous anomaly maps are binarized using thresholds calibrated on a validation set and fixed prior to final inference. Connected components with an area below 0.002% of the total image area are subsequently removed to suppress isolated noisy activations while retaining structurally coherent anomalous regions.

2.1.2. Architecture

The proposed architecture extends INP-Former [6] with three principal modifications: a DINOv3 Vision Transformer encoder for richer patch-level representations, a graph attention context module for local structural modeling after the INP bottleneck, and convolutional local-context refinement at every decoder level. The overall pipeline comprises a frozen vision transformer encoder, an INP extractor, an INP bottleneck, a graph attention context module, and an INP-guided decoder.

We adopt the DINOv3 Vision Transformer backbone [8] as the image encoder, using the ViT-L/16 configuration in our final model. Given an input image, the encoder extracts patch-level features from multiple intermediate transformer layers, each with a feature dimension of 1024. Features drawn from layers of varying depth are fused to yield multi-level representations that jointly encode low-level structural detail and high-level semantic content. The encoder is kept frozen throughout training to preserve its strong pretrained representations and to mitigate overfitting to the limited pool of normal training samples.

Following INP-Former, we maintain a set of learnable Intrinsic Normal Prototypes (INPs) that summarize the distribution of normal visual patterns. Let $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N}$ denote the fused patch tokens extracted by the encoder, where N is the total number of spatial patch tokens. A set of INPs, $\mathbf{P} = \{\mathbf{p}_m\}_{m=1}^{M}$, is updated by the INP extractor through cross-attention aggregation over $\mathbf{X}$. The extracted prototypes subsequently guide the decoder in reconstructing normal feature representations.

After the INP bottleneck, we introduce a Graph Attention Context Block to explicitly model local patch-to-patch relationships. Each patch token is treated as a graph node, with edges defined over a multi-radius local neighborhood encompassing self, adjacent, and diagonal patches. For

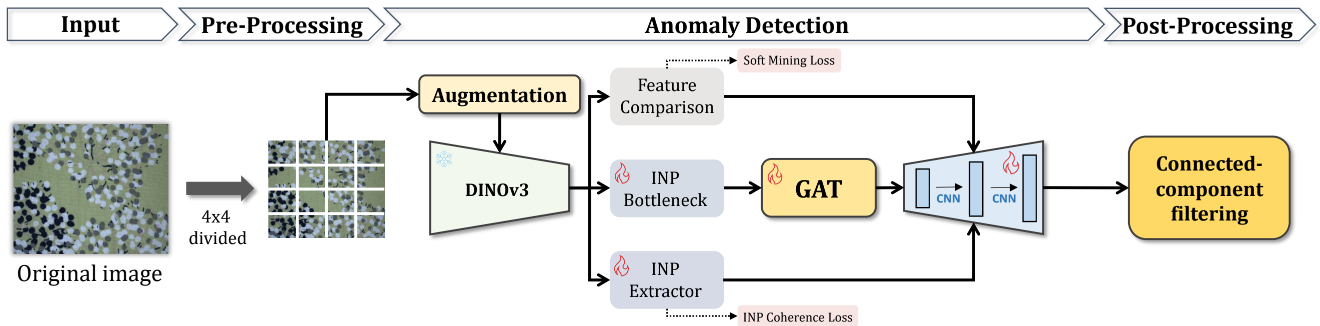

Figure 1. Overall architecture of the proposed framework. Input images are preprocessed via 4×4 grid splitting and fed into a frozen DINOv3 encoder. The extracted patch features pass through the INP extractor, INP bottleneck with graph attention context block, and INP-guided decoder with convolutional local-context refinement, yielding pixel-level anomaly maps that are finalized through postprocessing.

node i, the attention weight with respect to neighbor j is computed as

$$\alpha_{ij} = \frac{\exp\Big((W_q\mathbf{x}_i)^\top (W_k\mathbf{x}_j)/\sqrt{d}\Big)}{\sum_{j' \in \mathcal{N}(i)} \exp\Big((W_q\mathbf{x}_i)^\top (W_k\mathbf{x}_{j'})/\sqrt{d}\Big)}, \tag{1}$$

where $\mathcal{N}(i)$ denotes the local neighborhood of node i, and W_q, W_k, W_v are the query, key, and value projection matrices, respectively. The context-enhanced token is then obtained via a gated residual update:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i + \tanh(\gamma)\, W_o \sum_{j \in \mathcal{N}(i)} \alpha_{ij}\, W_v\mathbf{x}_j, \tag{2}$$

where W_o is the output projection and γ is a learnable gating scalar. The gated residual formulation stabilizes optimization and allows the model to adaptively regulate the contribution of graph-based contextual information.

The INP-guided decoder reconstructs normal feature representations by attending to the extracted prototypes $\mathbf{P}$. At each decoder level l, we apply a convolutional local-context block that reshapes the patch tokens into a two-dimensional feature map, applies a 3×3 convolutional branch for spatial refinement, and projects the result back to the token space via a gated residual:

$$\hat{\mathbf{z}}^l = \mathbf{z}^l + \tanh(\eta_l)\, \Phi_l\big(\mathbf{z}^l\big)\,, \tag{3}$$

where $\mathbf{z}^l$ denotes the decoder token at level l, $\Phi_l(\cdot)$ is the convolutional local-context module, and η_l is a level-wise learnable gating scalar. This mechanism improves local spatial consistency in the reconstructed features and helps preserve fine-grained normal structures around small or geometrically thin regions.

Pixel-level anomaly maps are generated by measuring the feature discrepancy between encoder and decoder representations. Let $\mathbf{E}^l$ and $\mathbf{D}^l$ denote the encoder and decoder feature maps at level l, respectively. The anomaly score at level l is defined as the cosine discrepancy:

$$\mathcal{A}^l = 1 - \frac{\mathbf{E}^l \cdot \mathbf{D}^l}{\|\mathbf{E}^l\|_2\, \|\mathbf{D}^l\|_2}. \tag{4}$$

Anomaly maps from all levels are aggregated and upsampled to the input image resolution. Since the INPs encode only normal patterns, anomalous regions—which cannot be faithfully reconstructed from the prototypes—yield elevated feature discrepancies, providing a reliable pixel-level indicator of abnormality.

2.1.3. Training

The model is trained under an unsupervised anomaly detection setting, using only normal images without any anomalous samples or defect annotations. Prior to training, each image is partitioned into a 4×4 grid, and the resulting sub-images are treated as independent training samples. This strategy increases the effective number of training samples and enables the model to learn fine-grained local normal structures at higher effective resolution. All input images are resized to 448×448. During training, we apply random horizontal flipping and brightness-based ColorJitter as data augmentation, followed by normalization using ImageNet mean and standard deviation. The brightness augmentation simulates illumination variation commonly encountered in industrial inspection environments, thereby improving robustness to lighting shifts at test time.

The model is trained for 10 epochs with a batch size of 16 and the number of INPs set to 6. We adopt StableAdamW as the optimizer with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}, paired with a warm cosine learning-rate scheduler that anneals the learning rate to 1×10^{-4} after the warm-up period. Gradient clipping is applied throughout to stabilize optimization. The trainable components include the INP extractor, INP bottleneck, INP tokens, INP-guided decoder, graph attention context module, decoder-level convolutional local-context modules, and feature-fusion weights, while the DINOv3 encoder remains frozen.

Following the INP-Former training objective, we optimize the model with a feature reconstruction loss and a prototype aggregation loss, and further introduce a prototype diversity regularization term to prevent the INPs from collapsing into degenerate representations. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + 0.2\,\mathcal{L}_{\text{gather}} + 0.01\,\mathcal{L}_{\text{div}}, \tag{5}$$

where $\mathcal{L}_{\text{rec}}$ measures the cosine discrepancy between encoder and decoder features, $\mathcal{L}_{\text{gather}}$ encourages normal patch tokens to align with the extracted INPs, and $\mathcal{L}_{\text{div}}$ promotes diversity among the prototype tokens. This objective preserves the prototype-guided reconstruction behavior of the base framework while enhancing the discriminative capacity of the learned normal representations.

During inference, the model produces a continuous anomaly map for each sub-image. Each map is smoothed with a Gaussian kernel, clipped to a valid range, and converted to a grayscale anomaly-score map. The sub-image maps are subsequently reassembled according to their grid indices to recover the full-image anomaly map, which is then passed to the postprocessing stage for thresholding and connected-component filtering.

2.2. Dataset and Evaluation

2.2.1. Data Utilization

We use the MVTec AD 2 dataset [4] for the Industrial Track of the VAND 4.0 Challenge. The dataset spans diverse industrial object categories captured under realistic conditions, including lighting variation, reflective and transparent materials, cluttered backgrounds, and small defects. These factors render pixel-level anomaly localization particularly challenging, as anomalous regions frequently occupy only a minor fraction of the total image area.

To better exploit the high-resolution imagery, we apply a uniform 4×4 grid-splitting strategy across all object categories and data subsets, including training, validation, public test, and private test images. Each image is decomposed into sixteen non-overlapping sub-images, which are processed independently by the model. This unified preprocessing preserves local structural detail and increases the effective resolution of small defect regions after resizing to the network input size.

During inference, each sub-image yields an independent anomaly map. To produce predictions in the original image space, the sub-image anomaly maps are reassembled into a full-resolution layout by grouping them according to their filename prefixes and grid indices, then placing each map into its corresponding spatial position. The reconstructed anomaly maps serve as the final continuous anomaly-score maps.

After binarization, we apply connected-component filtering to suppress spurious isolated responses. Let H and W denote the height and width of a binary mask. A connected component is discarded when its area falls below

$$A_{\min} = \max\big(8,\ 2 \times 10^{-5} HW\big)\,, \tag{6}$$

corresponding to 0.002% of the total image area, with an absolute minimum of 8 pixels. This step reduces false positive noise arising from isolated high-response pixels while retaining larger, spatially coherent regions that are more likely to correspond to genuine defects.

2.2.2. Evaluation Criteria

We follow the official evaluation protocol of the VAND 4.0 Industrial Track, in which the primary task is pixel-level anomaly segmentation on MVTec AD 2 and the principal metric is SegF1. SegF1 evaluates the quality of predicted binary anomaly masks at the pixel level via the harmonic mean of precision and recall:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{7}$$

where precision and recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{8}$$

and TP, FP, and FN denote true positive, false positive, and false negative pixels, respectively. A high SegF1 score indicates that the predicted masks accurately delineate true anomalous regions while minimizing spurious positive predictions.

Binary masks are obtained from continuous anomaly maps via a validation-calibrated fixed thresholding protocol, in which the threshold is determined prior to final inference and held constant throughout evaluation. The binarized masks are further refined by the connected-component filtering described above, suppressing noisy fragments that are unlikely to correspond to genuine defects.

In addition to the primary metric, we monitor a set of threshold-independent localization metrics during internal validation, including pixel-level AUROC, average precision, and AU-PRO. The per-region overlap at a given threshold is defined as

$$\text{PRO} = \frac{1}{K} \sum_{i=1}^{|D_{\text{test}}|} \sum_{j=1}^{k_i} \frac{\left|P_{\text{ano}}^{i} \cap C_{i,j}\right|}{|C_{i,j}|}, \tag{9}$$

where $C_{i,j}$ denotes the j-th ground-truth anomalous region in image i, P_{ano}^{i} denotes the predicted anomalous region, k_i is the number of anomalous regions in image i, and K is the total number of ground-truth anomalous regions across the test set. These auxiliary metrics serve as internal diagnostics of localization quality, while the final challenge ranking is governed by the official SegF1 protocol.

3. Results

3.1. Performance Metrics

We evaluate our method on the official VAND 4.0 Industrial Track protocol using the MVTec AD 2 dataset. The evaluation contains two private test sets: $TEST_{priv}$, which follows the regular lighting condition of the training set, and $TEST_{priv,mix}$, which contains both seen and unseen lighting conditions. Since the challenge ranking is based on pixel-level binary anomaly segmentation, we use segmentation F1 score (SegF1) as the primary metric. In addition, we report $AucPRO_{0.05}$ to evaluate the localization quality of continuous anomaly maps before final binarization.

Table 1 reports the category-wise $AucPRO_{0.05}$ scores of our method on $TEST_{priv}$ and $TEST_{priv,mix}$. Our method achieves **XX.XX%** mean AU-PRO$_{0.05}$ on $TEST_{priv}$ and **XX.XX%** mean $AucPRO_{0.05}$ on $TEST_{priv,mix}$. These results indicate that the proposed framework produces reliable continuous anomaly maps across diverse industrial objects. The $AucPRO_{0.05}$ gap between the two private splits is **XX.XX%**, showing the degree to which the model maintains localization quality under lighting variation.

Table 1. Segmentation $AucPRO_{0.05}$ Score (in %) of the Proposed Method on Binarized Images for $TEST_{priv}$ / $TEST_{priv,mix}$ Set

Object	$TEST_{priv}$	$TEST_{priv,mix}$
Can	-	-
Fabric	-	-
Fruit Jelly	-	-
Rice	-	-
Sheet Metal	-	-
Vial	-	-
Wall Plugs	-	-
Walnuts	-	-
Mean	-	-

3.1.1. Comparison

Table 2 summarizes the category-wise SegF1 scores on $TEST_{priv}$ / $TEST_{priv,mix}$. Our method achieves **XX.XX%** and **XX.XX%** mean SegF1 on the two splits, respectively, with an absolute gap of **XX.XX%**.

Compared with representative baselines, including PatchCore, RD, RD++, EfficientAD, MSFlow, SimpleNet, and DSR, our method achieves the best overall performance. It improves over the strongest baseline by **XX.XX** percentage points on $TEST_{priv}$ and **XX.XX** percentage points on $TEST_{priv,mix}$.

The improvement mainly comes from the combination of high-resolution local processing and context-aware feature reconstruction. The 4×4 grid splitting preserves small defect details, the frozen DINOv3 ViT-L/16 encoder provides robust patch-level features, and the Graph Attention Context Block with decoder-level Local Context Blocks improves local structural modeling. Finally, fixed thresholding and connected-component filtering convert continuous anomaly maps into stable binary masks for SegF1 evaluation.

4. Discussion

4.1. Challenges and Solutions

The main challenge of MVTec AD 2 is that anomalies are often very small compared with the full image resolution. Directly resizing the original image may suppress such fine defects and weaken pixel-level localization. To address this issue, we split each image into a fixed 4×4 grid and process the sub-images independently. This preserves local details after resizing and allows the model to detect subtle defects more effectively.

Another challenge is the lighting variation between training and test images, especially in $TEST_{priv,mix}$. Since the model is trained only with normal samples under regular conditions, unseen illumination can easily cause false positives. We mitigate this by using a frozen DINOv3 ViT-L/16 encoder and brightness-based ColorJitter during training. The DINOv3 encoder provides stable patch-level features, while ColorJitter improves robustness to photometric changes.

Finally, stable binarization is crucial because the official SegF1 metric evaluates binary masks. We therefore use a fixed threshold calibrated before final inference and apply connected-component filtering to remove tiny isolated responses. This reduces noisy false positives while preserving coherent anomalous regions.

4.2. Model Robustness and Adaptability

Our model is designed to remain robust across different object categories, lighting conditions, and defect scales. The frozen DINOv3 encoder reduces overfitting to the limited normal training set, and the INP-based reconstruction framework models normal feature patterns through intrinsic normal prototypes. When anomalous regions appear, they are not faithfully reconstructed from these prototypes, resulting in high feature discrepancy.

To further improve adaptability, we introduce the Graph Attention Context Block after the prototype bottleneck and Local Context Blocks in the decoder. These modules strengthen local structural modeling by allowing neighboring patch tokens to exchange contextual information and by restoring spatial inductive bias during reconstruction. As a result, the model can better handle diverse industrial patterns such as repetitive textures, reflective surfaces, transparent objects, and cluttered foregrounds without using

Table 2. Segmentation F1 (SegF1) score (in %) on binarized images for $TEST_{priv}$ / $TEST_{priv,mix}$ on the MVTec AD 2 dataset.

Object	Methods							
	Ours	PatchCore	RD	RD++	EfficientAD	MSFlow	SimpleNet	DSR
Can	**- / -**	0.3 / 0.1	0.1 / 0.1	0.1 / 0.1	0.8 / 0.1	5.0 / 0.1	0.6 / 0.1	0.4 / 0.1
Fabric	**- / -**	11.5 / 9.8	2.6 / 2.2	2.9 / 2.3	7.6 / 1.0	22.0 / 4.1	21.6 / 10.2	7.9 / 5.0
Fruit Jelly	**- / -**	8.7 / 8.2	22.5 / 22.7	26.9 / 26.7	20.8 / 18.2	47.6 / 38.1	25.1 / 23.0	17.9 / 17.2
Rice	**- / -**	3.8 / 4.2	7.0 / 3.9	9.5 / 2.9	15.0 / 0.5	19.1 / 1.8	11.6 / 1.0	1.5 / 1.4
Sheet Metal	**- / -**	1.8 / 1.1	41.3 / 39.2	40.9 / 37.7	9.3 / 3.8	13.0 / 7.6	14.6 / 2.8	13.9 / 14.9
Vial	**- / -**	2.3 / 2.2	28.0 / 28.3	28.2 / 22.8	30.5 / 26.5	23.3 / 6.2	31.9 / 17.5	28.2 / 27.9
Wall Plugs	**- / -**	0.0 / 0.0	1.9 / 0.8	1.3 / 0.9	4.4 / 0.3	0.1 / 0.2	1.0 / 0.3	0.4 / 0.4
Walnuts	**- / -**	1.2 / 1.3	41.2 / 36.7	44.1 / 40.5	34.6 / 13.3	44.5 / 14.3	35.2 / 14.3	17.0 / 9.6
Mean	**- / -**	3.7 / 3.4	18.1 / 16.7	19.2 / 16.7	15.4 / 8.0	21.8 / 9.0	17.7 / 8.7	10.9 / 9.5

category-specific architectures.

4.3. Future Work

Although the proposed method achieves strong anomaly segmentation performance, several improvements remain possible. First, the current 4×4 grid splitting strategy uses non-overlapping crops, so anomalies near grid boundaries may be fragmented. Future work could explore overlapping crops or boundary-aware merging to improve mask continuity.

Second, our final prediction still relies on a fixed threshold. Developing an adaptive thresholding strategy that works across different categories, lighting conditions, and defect scales could further improve SegF1. Finally, since grid splitting increases the number of forward passes per image, future work could investigate lightweight encoders, token pruning, or knowledge distillation to reduce inference cost while maintaining localization accuracy.

5. Conclusion

In conclusion, our solution for the VAND 4.0 Industrial Track addresses the main challenges of the MVTec AD 2 dataset, including high-resolution inputs, small defects, and lighting variations. The proposed framework is based on prototype-guided feature reconstruction and adopts a frozen DINOv3 ViT-L/16 encoder for robust patch-level representation. We further introduce a Graph Attention Context Block for local structural modeling and decoder-level Local Context Blocks for spatially consistent feature reconstruction. In addition, the 4×4 grid splitting strategy preserves fine local details, while fixed thresholding and connected-component filtering generate stable binary anomaly masks.

Our method achieves **XX.XX% / XX.XX%** SegF1 and **XX.XX% / XX.XX%** $AucPRO_{0.05}$ on $TEST_{priv}$ / $TEST_{priv,mix}$, respectively. These results demonstrate that combining strong self-supervised visual features, prototype-based reconstruction, and explicit local-context modeling is effective for robust industrial anomaly segmentation. Overall, our solution provides a practical class-agnostic pipeline for detecting and localizing defects without using anomalous training samples.

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